

SUVRADIP GHOSH

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EDUCATION

Master of Computer Applications	2024 – Present
Sikkim Manipal Institute of Technology	CGPA: 8.25/10
Bachelor of Computer Applications	2021 – 2024
Techno India University	CGPA: 8.39/10
Higher Secondary (Class XII)	2021
Maynaguri High School	88.6%

TECHNICAL SKILLS

Programming Languages	Python, Java, JavaScript, C
Frontend	React.js, HTML, CSS, Tailwind CSS, Redux, Zustand
Backend	Node.js, Express.js, FastAPI
Databases	MongoDB, SQLite, Redis
AI / ML	PyTorch, TensorFlow, Scikit-Learn, OpenCV, SHAP
Tools	Git, Docker, Clerk, Convex, ImageKit

PROJECTS

InsightATS | AI Hiring Support Platform with Project Evaluation

[Live](#) | [GitHub](#)

Tech Stack: React.js • Redux • FastAPI • PyTorch • Redis • BERT • RoBERTa • DistilBERT • SHAP

- Built an AI-powered hiring web-platform integrating **Project Complexity Evaluation** alongside **Contextual Resume-JD Matching** and resume parsing to rank candidates beyond keyword-based ATS screening.
- Fine-tuned and integrated BERT, RoBERTa, and DistilBERT models, achieving **0.75 F1** and **0.78, 0.93 Macro F1** for extraction, resume-jd matching, and project classification tasks.
- Reduced inference latency by 51%** (6.62s → 3.24s) using FastAPI lazy loading, in-memory model orchestration, and Redis caching.
- Reduced client-to-server API requests by 90%** through shared job-description batch processing while Redux managed asynchronous file uploads and application state for candidate screening.

CodeNest | AI-Powered Cloud IDE

[Live](#) | [GitHub](#)

Tech Stack: React.js • Zustand • Express.js • Docker • MongoDB Atlas • Redis • Groq Llama-3.3-70B

- Built a cloud IDE supporting **20+ concurrent users**, featuring **isolated Docker containerized secure code execution**, AI code reviews, flowchart visualization, and **5-second** execution timeouts.
- Engineered an asynchronous execution pipeline using a **25-job queue** and worker processes to improve reliability under concurrent workloads and prevent resource exhaustion.
- Reduced AI response latency to 1–3 seconds** by integrating **Groq Llama-3.3-70B** with Redis-backed caching, in-memory fallback, and secure API rate limiting using SHA-256 cache keys.
- Optimized state management** by migrating authentication, project, and version state to persistent **Zustand stores**, eliminating React Context nesting.

Skin Disease Detection using CNN

[Kaggle](#) | [GitHub](#)

Tech Stack: Python • TensorFlow • EfficientNetB3 • OpenCV • Scikit-Learn

- Developed a **deep learning system to recognize skin diseases** using the **HAM10000** dataset (10,015 dermatoscopic images), categorizing lesions into benign, melanoma, and non-melanoma classes.
- Improved model reliability through patient-wise data splitting to eliminate data leakage, **CLAHE**-based image enhancement, hair-removal preprocessing, and **transfer learning with EfficientNetB3**.
- Achieved **90% test accuracy** with F1 scores of **0.89 (Benign)**, **0.86 (Melanoma)**, and **0.94 (Non-Melanoma Cancer)** using focal loss, transfer learning, and test-time augmentation.

CERTIFICATIONS

Android Developer Virtual Internship (Google & AICTE EduSkills)	Apr 2024 – Jun 2024
Grade: Excellent	Credential
IBM Professional Certificate: Full Stack Software Developer	Credential
IBM Professional Certificate: Data Science and AI	Credential